Tenchi Sales & Operations Planning

Strategic Business Planning

March’2026

Strategy. Scale. Success.

Forecasting Automation Model for S&OP

An automated S&OP forecasting engine transforms demand planning from reactive spreadsheets into a structured, data-driven decision system.

By combining real-time inventory visibility with AI-driven demand forecasting, organizations can proactively align sales expectations with supply capabilities.

The result is **higher service levels, optimized inventory investment, and stronger operational control across the entire supply chain.**

Objective

Automate forecasting and inventory planning using key operational inputs to ensure **optimal stock availability, lower inventory holding costs, and improved service levels**.

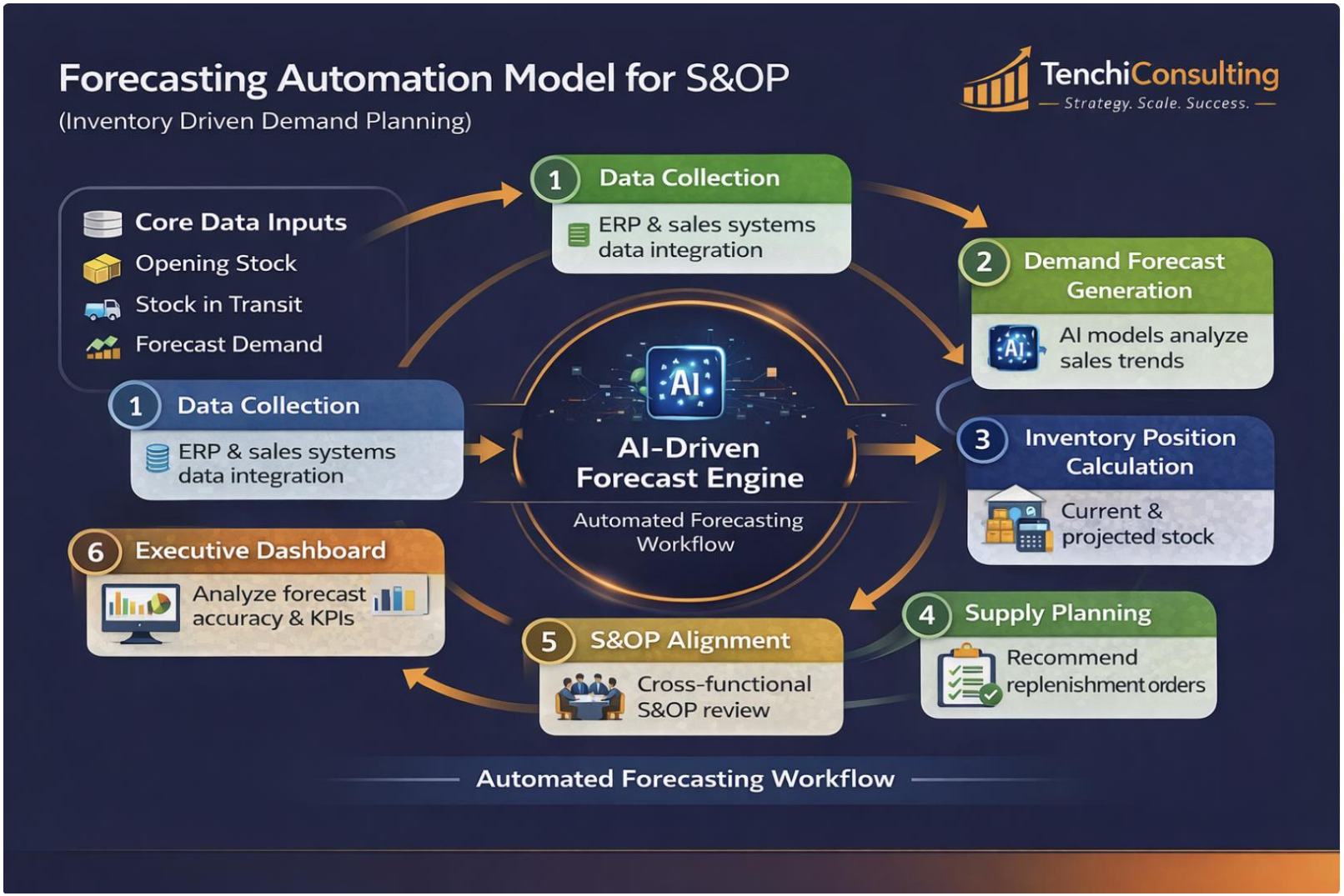
Core Data Inputs Required :

Data Input	Unit	Description
Opening Stock	Cartons	Inventory available at beginning of month
Past Month Actual Sales	Cartons	Historical demand data
Stock in Transit	Cartons	Inventory shipped but not yet received
Forecasted Demand	Cartons	AI / statistical forecast based on past months avg
Safety Stock	Cartons	Buffer inventory
Replenishment Quantity	Cartons	Planned supply order

Closing Stock = Opening Stock + Stock in Transit – Actual Sales

Replenishment Quantity = (Forecast Demand + Safety Stock) – Closing Stock

WorkFlow


Forecasting Automation Model for S&OP
(Inventory Driven Demand Planning)
TenchiConsulting
Strategy. Scale. Success.
Core Data Inputs
Opening Stock
Stock in Transit
Forecast Demand
1 Data Collection
ERP & sales systems data integration
2 Demand Forecast Generation
AI models analyze sales trends
3 Inventory Position Calculation
Current & projected stock
4 Supply Planning
Recommend replenishment orders
5 S&OP Alignment
Cross-functional S&OP review
6 Executive Dashboard
Analyze forecast accuracy & KPIs
AI
AI-Driven Forecast Engine
Automated Forecasting Workflow
1 Data Collection
ERP & sales systems data integration
Automated Forecasting Workflow

Dashboard KPIs for Management

KPI	Description
Forecast Accuracy	% deviation vs actual sales
Inventory Turns	Efficiency of inventory usage
Fill Rate	Customer order fulfillment
Stock Coverage	Days of inventory available
Stockout Risk	Probability of shortages

Benefits

Operational Benefits :

Improved demand planning accuracy
Reduced stockouts
Lower inventory carrying cost

Financial Benefits :

Better working capital management
Reduced excess inventory
Higher revenue realization

Strategic Benefits:

Data-driven S&OP decisions
Improved cross-functional alignment
Scalable demand planning model

Thank you !